



3rd Round

Haramaya University
Haramaya Institute of Technology
Department of Mechanical Engineering

***Exit Examination for Prospective BSc Degree Graduate in
Mechanical Engineering***

Time Allowed: 3 hrs.

General Instruction

- ✍ Please write your details(name & ID) on the answer sheet space provided;*
- ✍ Make sure the total number of questions is 124;*
- ✍ Read the questions carefully, choose your correct answer from the given alternatives, and shade the correct answer on the separately attached answer sheet form;*
- ✍ Cheating or any attempt of copying from others during the exam is strictly prohibited;*
- ✍ Using a mobile phone is strictly forbidden;*
- ✍ Using an unnecessary piece of paper or any form of written thing on different body parts lead to disqualification from the exam; and*
- ✍ At the end of the exam, the question paper will be returned along with the answer sheet.*

Prepared by
Department of Mechanical Engineering
June 21, 2023

1. Pick up the wrong statement Nickel and chromium in steel help in

- A. Providing corrosion resistance
- B. Improving machining properties
- C. Providing high strength at elevated temperatures
- D. Raising the elastic limit

Answer: B

2. The number of atoms per unit cell for an FCC crystal is

- A. 2
- B. 4
- C. 6
- D. 8

Answer: B

3. Which of the following is the surface defect?

- A. Vacancy
- B. Screw dislocation
- C. Twin Boundaries
- D. Edge Dislocation

Answer: C

4. Which proportion of the creep curve provides the information on the steady-state creep rate?

- A. Primary stage
- B. Secondary stage
- C. Tertiary stage
- D. All of the above

Answer: B

5. Izod and Charpy tests are used to measure of the materials.

- A. Tensile Strength
- B. Compressive
- C. Toughness
- D. Hardness

Answer: C

6. In a phase diagram, the line above which the phases are completely liquids is called.

- A. Solidus
- B. Liquidus
- C. Eutectic
- D. Peritectic

Answer: B

7. Phase transformation in metals /alloys occurs by

- A. Random motion
- B. Explosion
- C. Rotation and Vibration
- D. Nucleation and Growth

Answer: D

8. Cyaniding and Nitriding are two methods of

- A. Normalizing
- B. Hardening
- C. Tempering
- D. Case hardening

Answer: D

9. Solvent extraction is basically known as

- A. Gas Liquid Extraction
- B. Liquid-Liquid Extraction
- C. Liquid – Solid extraction
- D. None of the mentioned

Answer: B

10. The metal used as in matrix in metal matrix composites

- A. Nickel
- B. Lead
- C. Boron
- D. Titanium

Answer: D

11. A bar of length 3m is tapered uniformly from diameter 0.16m to 0.08m and subjected to 10N axial load. Then the extension of the bar will be [Take $E = 2 \times 10^5 \text{ N/m}^2$]

- A. 0.015m
- B. 0.047m
- C. 0.147m
- D. 0.009m

Answer: A

12. Poisson's ratio is the ratio of

- A. Lateral stress to longitudinal stress
- B. Longitudinal stress to lateral stress
- C. Lateral strain to longitudinal strain
- D. Longitudinal strain to lateral strain

Answer: C

13. Find the strain energy stored in the steel rod of 2m long and 0.06m diameter when a tensile load of 60KN is gradually applied take $E = 10^{11} \text{ N/m}^2$.

- A. 25.46Nm
- B. 12.73Nm
- C. 6.37Nm
- D. 3.18Nm

Answer: B

14. A cantilever beam of length L is subjected to a moment M at the free end. Then the maximum deflection occurs at

- A. Free end
- B. Fixed end
- C. The middle
- D. Any point

Answer: A

15. A cylindrical pipe of diameter 30cm and thickness 1.2 cm is subjected to an internal fluid pressure of 9 KN/m^2 then the maximum shear stress developed be

- A. 14.06 KN/m^2
- B. 112.5 KN/m^2
- C. 56.25 KN/m^2
- D. 28.13 KN/m^2

Answer: D

16. Column is defined as

- A. Horizontal member of structure which carries a tensile load
- B. Vertical member of a structure that carries a tensile load
- C. Vertical member of a structure that carries an axial compressive load
- D. Horizontal member of a structure that carries no load

Answer: C

17. Find the torque transmitted by a solid shaft of 0.2m diameter if the maximum shear stress induced is 40KN/m^2

- A. 12.57Nm
- B. 62.83Nm
- C. 31.42Nm
- D. 15.71Nm

Answer: B

18. A rectangular bar of uniform cross-sectional area is subjected to a tensile stress σ_1 in the radial direction and compressive stress σ_2 in the axial direction. Then on the plane of maximum shear stress of the bar, there will be

- A. A maximum normal stress
- B. Minimum normal stress
- C. No normal stress
- D. Normal and shear stress

Answer: D

19. A torque of 8Nm is applied to a hollow shaft having a rectangular cross-sectional area of 1.2m^2 and thickness of 0.08m . The shear flow through the wall of the shaft be

- A. 3.33KN/m
- B. 41.61KN/m
- C. 6.67KN/m
- D. 50KN/m

Answer: A

20. A thick cylinder is subjected to internal fluid pressure and develops a principal stress in the wall of the cylinder. Which stress type is not developed inside the cylinder due to the applied pressure

- A. Hoop tensile stress
- B. Longitudinal tensile stress
- C. Radial tensile stress
- D. Circumferential tensile stress

Answer: C

21. A propped cantilever beam of length L supports a vertical uniform load of intensity q . then the number of unknown reactions in the beam will be

- A. 4
- B. 3
- C. 2
- D. 5

Answer: B

22. A curved beam of circular cross-section of diameter 30mm is subjected to a pure bending moment of 100KNM and the radius of curvature is 90mm . find the location of the neutral axis. Take the constant $h^2=500$

- A. -10.71mm
- B. 5.23mm
- C. 10.71mm
- D. -5.23mm

Answer: D

23. Which one of the following is true?

- A. In solids, conduction heat transfer is due to the collisions and diffusion of the molecules during their random motion.
- B. In gases and liquids, conduction heat transfer is due to the combination of vibrations of the molecules in a lattice and the energy transport by free electrons.
- C. The faster the fluid motion, the greater the convection heat transfer.
- D. The net exchange of heat between the two radiating surfaces is due to the fact that one at the higher temperature radiates more and receives high energy for its absorption.

Answer: C

24. During a heat treatment process, spherical balls of 12 mm diameter are initially heated to 800°C. Then they are cooled to 100°C, by immersing them in an oil bath of 35°C, with a convection coefficient of 20 W/m²°C. Determine the time required for the cooling process. Thermo-physical properties of the balls are k=50 W/m°C, $\rho = 7750 \text{ kg/m}^3$, C = 520 kJ/kg°C.

- | | |
|-------------------|-------------------|
| A. 919.95 seconds | C. 928.75 seconds |
| B. 993.95 seconds | D. 969.75 seconds |

Answer: B

25. All are true to increase the convective heat transfer except;

- A. Increase the temperature difference between the surface and the fluid.
- B. Increase the fluid flow velocity.
- C. Increase the convection coefficient
- D. Decrease the contact surface area.

Answer: D

26. Which one of the following is true about the effectiveness of the fin in heat transfer;

- A. If the fin is used to improve heat dissipation from the surface, then the fin effectiveness must be greater than unity.
- B. To improve the effectiveness of the fin, the fin should be made from high-conductive material
- C. A high value of film coefficient has an adverse effect on effectiveness.
- D. Effectiveness of the fin can also be increased by decreasing the ratio of the perimeter to the cross-sectional area.

Answer: D

27. In the following events; which one is subjected to periodic variation;
- A. Heat processing of regenerators whose packing's are alternately heated by flue gases and cooled by air
 - B. Heating or cooling of an ingot in a furnace
 - C. Daily surface temperature of the earth.
 - D. Cooling of bars, blanks, and metal billets in steelworks

Answer: C

28. Which one of the following dimensional analyses is not correct;

Where; $M = \text{mass}$, $L = \text{length}$, $T = \text{time}$ and $\theta = \text{temperature}$

- A. Fluid viscosity = $M^1 L^{-1} \theta^{-1}$
- B. Heat transfer coefficient = $M^1 T^{-3} \theta^{-1}$
- C. Thermal conductivity = $M^1 L^1 T^{-3} \theta^{-1}$
- D. Specific heat = $L^2 T^{-2} \theta^{-1}$

Answer: A

29. The pressure of the liquid flowing through the divergent portion of a venture meter is

- A. Remains constant
- B. Decreases
- C. Increases
- D. Depends upon the mass of liquid

Answer: B

30. Bulk modulus of a fluid is the ratio of

- A. Shear stress to shear strain
- B. Increase in volume to the viscosity of the fluid
- C. Increase in pressure to the volumetric strain
- D. Critical velocity to the viscosity of the fluid

Answer: C

31. Which of the following is **NOT** a type of force considered in the Navier-Stokes equation?

- A. Gravity force
- B. Pressure force
- C. Surface tension force
- D. Viscous force

Answer: C

32. Which among the following is **NOT** a criterion to achieve similitude?

- A. Geometric similarity
- B. Kinematic similarity
- C. Dynamic similarity
- D. Conditional similarity

Answer: D

33. Two pipes, each of diameter d , converge to form a pipe of diameter D . What should be the relation between d and D such that the flow velocity in the third pipe becomes double that in each of the two pipes?

- A. $D = d$
- B. $D = 2d$
- C. $D = 3d$
- D. $D = 4d$

Answer: A

34. Which of the following is **CORRECT**

- A. Pathlines of two particles in a one-dimensional flow can never intersect
- B. Pathlines of two particles in a one-dimensional flow can never intersect if the two particles move along the same direction
- C. Pathlines of two particles in a one-dimensional flow can intersect only if the two particles move along different directions
- D. Pathlines of two particles in a one-dimensional flow can intersect only if the two particles move along the same direction

Answer: D

35. In a mechanical refrigeration system, the highest temperature of refrigerant occurs

- A. Between compressor and condenser
- B. Before the expansion valve
- C. In evaporator
- D. Between condenser and evaporator

Answer: A

36. The vapor compression refrigerator employs the following cycle

- A. Rankine
- B. Carnot
- C. Reversed Rankine
- D. Reversed Carnot

Answer: D

37. The ratio of sensible heat to total heat is known as

- A. Dew point
- B. Humidification ratio
- C. Relative humidity
- D. Sensible heat factor

Answer: D

38. The most common type of absorption system used in industrial applications is based on the refrigerant absorbent combination of

- A. Air water
- B. Lithium bromide air
- C. Carbon dioxide air
- D. Ammonia water

Answer: D

39. In the central air conditioning system

- A. Entire equipment is placed in the air-conditioned space
- B. There is only one machine that performs the function of air conditioning
- C. All the major equipment is located in one central space removed from the air to be conditioned
- D. Is used when a small area is air-conditioned

Answer: C

40. A device designed to remove moisture from a refrigerant is called
- A. Expansion valve
 - B. Dehumidifier
 - C. Drier
 - D. Solenoid

Answer: C

41. The sum of all the microscopic forms of energy is called
- A. Sensible energy
 - B. Latent energy
 - C. Internal energy
 - D. Chemical energy

Answer: C

42. At the Adama site evaluated a wind farm is observed to have steady winds at a speed of 8.5 m/s for a mass of 10 kg . Determine the wind energy.
- A. 361.25 J
 - B. 85 J
 - C. 850 J
 - D. 425 J

Answer: A

43. Determine the power required for an 1150-kg car to climb a 100-m -long uphill road with a slope of 30° (from horizontal) in 12 s at a constant velocity?
- A. 690 kW
 - B. 47.06 kW
 - C. 2.395 kW
 - D. 81.33 kW

Answer: B

44. Water is heated in a closed pan on top of a range while being stirred by a paddle wheel. During the process, 30 kJ of heat is transferred to the water, and 5 kJ of heat is lost to the surrounding air. The paddle-wheel work amounts to 500 Nm . Determine the final energy of the system if its initial energy is 10 kJ .
- A. 15.5 kJ
 - B. 25.5 kJ
 - C. 10.5 kJ
 - D. 35.5 kJ

Answer: D

45. A Haramaya University campus has 400 classrooms and 300 faculty offices. The classrooms are equipped with 8 fluorescent tubes, each consuming 120 W , including the electricity used by the ballasts. The faculty offices, on average, have half as many tubes. The campus is open 240 days a year. The classrooms and faculty offices are not occupied for an average of 4 h a day, but the lights are kept on. If the unit cost of electricity is $\$0.12/\text{kWh}$, determine how much the campus will save a year if the lights in the classrooms and faculty offices are turned off during unoccupied periods.
- A. $\$60,825.6/\text{year}$
 - B. $\$55,757/\text{year}$
 - C. $\$67,556/\text{year}$
 - D. $\$57,678/\text{year}$

Answer: A

46. A rigid tank contains air at 500 kPa and 150°C. As a result of heat transfer to the surroundings, the temperature and pressure inside the tank drop to 65°C and 400 kPa, respectively. Determine the boundary work done during this process.

- A. 100 kJ
- B. 10 kJ
- C. 15 kJ
- D. 0 kJ

Answer: D

47. Air at 20°C and 90 kPa enters the diffuser of a jet engine steadily with a velocity of 200 m/s. The inlet area of the diffuser is 0.4m^2 . The air leaves the diffuser with a velocity that is very small compared with the inlet velocity. Determine the mass flow rate of the air.

Use gas constant $R=0.287\text{kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$.

- A. 78.8 kg/s
- B. 86.5 kg/s
- C. 85.6 kg/s
- D. 79.8 kg/s

Answer: C

48. Air at 100 kPa and 290 K is compressed in a compressor steadily to 600 kPa and 380 K. The mass flow rate of the air is 0.02 kg/s, and a heat loss of 16 kJ/kg occurs during the process. Assuming the changes in kinetic and potential energies are negligible, determine the necessary power input to the compressor. Use enthalpy $h_{@290\text{K}} = 290.16\text{kJ/kg}$ and $h_{@380\text{K}} = 380.77\text{kJ/kg}$

- A. 2.74 kW
- B. 2.18 kW
- C. 2.81 kW
- D. 2.47 kW

Answer: B

49. Heat is transferred to a heat engine from a furnace at a rate of 90 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the thermal efficiency of this heat engine.

- A. 0.44
- B. 0.36
- C. 0.55
- D. 0.27

Answer: A

50. _____ is the ratio of the total amount of heat removed by an air conditioner or heat pump during a normal cooling season (in Btu) to the total amount of electricity consumed (in watt-hours, Wh).

- A. Energy efficiency ratio
- B. Coefficient of performance
- C. Thermal efficiency
- D. Seasonal energy efficiency

Answer: D

51. Among from the given factors that cause a process to be irreversible is it not the factor that causes a process to be irreversible?

- A. Unrestrained expansion
- B. Mixing of two fluids
- C. Electric resistance
- D. Pressure

Answer: D

52. Clausius inequality expression is

A. $\oint \frac{\delta Q}{T} \leq 1$
B. $\oint \frac{\delta Q}{T} \leq 0$

C. $\oint \frac{\delta Q}{T} \geq 1$
D. $\oint \frac{\delta Q}{T} \geq 0$

Answer: B

53. ----- type of instrument has identical calibration for AC as well as DC values

- A. Hotwire
B. Moving coil

- C. Induction
D. Moving iron

Answer: A

54. The use of ----- instruments is merely confined within laboratories as standardizing instruments.

- A. Absolute
B. Indicating

- C. Recording
D. Integrating

Answer: A

55. Excitation and amplification systems are needed for:

- A. For active transducers only
B. For passive transducers only
C. For both active and passive transducers
D. For both passive and output transducers

Answer: D

56. An Ideal OP-AMP has gain of -100. The input is connected to inverting end and the input resistance is $1\text{K}\Omega$. The feedback resistance is,

- A. $100\text{K}\Omega$
B. 10Ω

- C. 100Ω
D. $1000\text{K}\Omega$

Answer: D

57. A differential amplifier has +100 mv applied to the noninverting end and +250 mv applied to the inverting end. The output is 1.5 v the gain of the amplifier is:

- A. 10
B. 4.29

- C. 0.6
D. 15

Answer: A

58. A potentiometer is basically a:

- A. Deflection-type instrument
B. Null-type instrument
C. Deflection as well as null type instrument
D. A digital instrument

Answer: B

59. Digital to Analog Conversion is _____ Analog to Digital Conversion.

- A. Less complex than
- B. More Complex than
- C. As complex as
- D. Unpredictable

Answer: A

60. Which of the following is/are characteristic/s of mechatronic products and systems?

- A. Functional interaction between mechanical, electronic, and information technologies
- B. Special interaction of subsystems in one physical unit
- C. Intelligence related to the control functions of the mechatronics system
- D. All of the above

Answer: D

61. What is the function of an input signal conditioning unit?

- A. To produce control signals
- B. To amplify the signal and convert it into digital form
- C. To perform Mechanical work
- D. To produce electrical signals

Answer: B

62. Arrange the following components of the temperature measurement system according to the measurement systems hot body, Display system, Thermocouple, amplifier, and converter.

- A. Hot body---Display system----Thermocouple-----Amplifier and converter
- B. Hot body---Amplifier and converter-----Thermocouple-----Display system
- C. Hot body---Thermocouple-----Amplifier and converter-----Display system
- D. Display system-----Hot body---Thermocouple-----Amplifier and converter

Answer: C

63. What is the code for mechatronics engineering?

- A. CIP Code 15.4301
- B. CIP Code 14.4501
- C. CIP Code 14.4201
- D. CIP Code 15.4011

Answer: C

64. The device that generates the basic timing clock signal for the operation of the circuit using a crystal oscillator is

- A. Timing unit
- B. Timing and control unit
- C. Oscillator
- D. Clock generator

Answer: C

65. If the engine coolant leaks into the engine oil, then the engine oil
- A. Appears milky
 - B. Becomes foamy
 - C. Turns black
 - D. Turns to muddy

Answer: A

66. A clutch is usually designed to transmit maximum torque which is
- A. Equal to the maximum engine torque
 - B. 80 per cent of the maximum engine torque
 - C. 150 per cent of the maximum engine torque
 - D. 100 per cent of the maximum engine torque

Answer: C

67. The main function of the intake manifold is that it
- A. Promotes the mixture of air and fuel
 - B. Reduces intake noise
 - C. Cools the intake air to a suitable temperature
 - D. Distributes intake air equally to the cylinders

Answer: D

68. The firing order for an in-line four-cylinder I.C. engine is
- A. 1-2-3-4
 - B. 1-3-4-2
 - C. 1-2-4-3
 - D. 1-3-2-4

Answer: B

69. If the air-fuel mixture in a spark ignition engine is too rich, then the air-fuel ratio is about
- A. 17:1
 - B. 15:1
 - C. 13:1
 - D. 10:1

Answer: D

70. The diagram which shows the correct crank positions corresponding to the opening and closing of the valves, is known as
- A. Indicator diagram
 - B. Axial force diagram
 - C. Valve timing diagram
 - D. PV Diagram

Answer: C

71. The shear strength, tensile strength and, compressive strength of a rivet joint are 100 N, 120 N and 150 N respectively. If strength of the unriveted plate is 200 N, the efficiency of rivet joint is:
- A. 60%
 - B. 75%
 - C. 80%
 - D. 50%

Answer: D

72. A cotter joint is used to connect rods which are in:

- A. Compression only
- B. Tension and compression only
- C. Shear
- D. Tension only

Answer: B

73. A backing ring is used inside the pipe joint when making a

- A. Butt weld
- B. Fillet weld
- C. Sleeve weld
- D. Socket weld

Answer: A

74. The distance between the centres' of the rivets in adjacent rows of zigzag riveted joint is known as

- A. Pitch
- B. Back pitch
- C. Diagonal pitch
- D. Diametric pitch

Answer: C

75. The crest diameter of a screw thread is same as

- A. Major diameter
- B. Minor diameter
- C. Pitch diameter
- D. Core diameter

Answer: A

76. In the flange coupling the two flanges are coupled together by means of bolts fitted in

- A. Reamed holes
- B. Machined holes
- C. Threaded holes
- D. Gasketed holes

Answer: A

77. Lewis's equation in spur gear design applied to:

- A. Gear
- B. Stronger of the pinion or gear
- C. Weaker of the pinion or gear
- D. Pinion

Answer: C

78. The inner and outer radius of friction surface of a plate clutch are 50 mm and 100 mm respectively. What is the ratio of maximum intensity of pressure to minimum intensity of pressure on clutch plate if magnitude of axial force is 4 kN? Assume uniform wear theory.

- A. 4
- B. 2
- C. 6
- D. 8

Answer: B

79. The most suitable bearing for carrying very heavy loads with slow speed is

- A. Hydrodynamic bearing
- B. Ball bearing
- C. Roller bearing
- D. Hydrostatic bearing

Answer: D

80. What is the principle on which brakes work?

- A. Vibration
- B. Friction
- C. Suction
- D. Dragging

Answer: B

81. Which of the following is antifriction bearing?

- A. Needle bearing
- B. Pedestal bearing
- C. Collar bearing
- D. Hydrostatic bearing

Answer: A

82. Which of the following joins two rotating shafts to each other?

- A. Key
- B. Coupling
- C. Gear
- D. Belt drive

Answer: B

83. In air compressor performance curve, a surge line represents

- A. Limit of compressor discharge
- B. Limit of compressor efficiency
- C. Limit of stable operation
- D. Lower critical speed of shaft

Answer: C

84. From the following turbo-machine element which one is acting under low specific speed?

- A. Gas Turbine
- B. Kaplan Turbine
- C. Pelton Turbine
- D. Wind Turbine

Answer: C

85. The dimensions of Energy Per unit Mass are ____ (Where; M = mass, L = length, T = time)

- A. ML^2T^2
- B. $ML^{-2}T^{-2}$
- C. $L^{-2}T^{-2}$
- D. L^2T^{-2}

Answer: D

86. For Performance Characteristics of Hydraulic Turbines which one is wrong matched?

- A. Main = constant head characteristic curves.
- B. Operating = constant speed characteristics curves
- C. Overall efficiency = Muschel curves
- D. Constant efficiency = Iso-efficiency curve

Answer: C

87. If speed of Centrifugal Pump is increase, its NPSH requirements.

- A. Decrease
- B. Increase
- C. Remains Constant
- D. It's independent of Speed

Answer: B

88. In Turbo-Machinery rotary design, which one is wrongly formulated?

- A. Impulse Turbine Jet ratio $(m) = \frac{D}{d}$
- B. Impulse Turbine Number of buckets on a runner $(z) = 15 + \frac{D}{2d}$
- C. Reaction Turbine Area of flow at inlet $= \frac{\pi}{4} (D_{O2} - D_{b2})$
- D. Overall efficiency $(\eta_0) = \frac{QgH\rho}{Shaft\ Power}$

Answer: D

89. Let's consider a slider, which moves on a fixed link with curved surface. Thus, where does the instantaneous center locate?

- A. At the center of curvature
- B. On their point of contact
- C. At the center of circle
- D. At the pin joint

Answer: A

90. Which of the following displacement diagrams should be chosen for better dynamic performance of a cam-follower mechanism?

- A. Parabolic motion
- B. Simple harmonic motion
- C. Cycloidal motion
- D. Hyperbolic motion

Answer: C

91. Consider a motor vehicle moving at a certain speed takes a left turn in a curved path. If the engine rotates in the same direction as that of the wheels, then due to the centrifugal forces;

- A. The reaction on the inner wheels increases and on the outer wheels decreases
- B. The reaction on the outer wheels increases and on the inner wheels decreases
- C. The reaction on the front wheels increases and on the rear wheels decreases
- D. The reaction on the rear wheels increases and on the front wheels decreases

Answer: B

92. For which of the following mechanism does the Coriolis component of acceleration considered?

- A. Quick return motion mechanism
 - B. Four bar chain mechanism
 - C. Crank mechanism
 - D. Toggle mechanism
- slider

Answer: A

93. Which of the following condition describes differential gear box in automobiles?

- A. Assisting turning condition
- B. Speed reduction
- C. Assisting speed changing
- D. provide jerk-free movement

Answer: A

94. Which of the following cam follower is most applicable in automobile engines?

- A. Spherical faced follower
- B. Knife edge follower
- C. Flat faced follower
- D. Roller follower

Answer: A

95. Chains for material handling equipment are generally made of

- A. Carbon steel
- B. Cast iron
- C. Wrought iron
- D. Mild steel

Answer: A

96. Which one of the following shovel excavators is considered most efficient in loading carriers?

- A. Clam shell
- B. Back hoe
- C. Dipper shovel
- D. Drag line

Answer: C

97. Pneumatic tyres are not used for providing

- A. Friction
- B. Traction
- C. Comfort
- D. Speed

Answer: A

98. The number of cylinders in case of steam locomotive is

- A. One
- B. Two
- C. Four
- D. Eight

Answer: B

99. Which of the following cannot be handled using lifting magnets?

- A. Castings of cast iron
- B. High manganese steel
- C. Mild steel plates
- D. Scarp iron

Answer: B

100. In case of belt conveyors, the bearings used for return idlers are

- A. Anti-friction bearings
- B. Bush bearings
- C. Cast iron bearings
- D. Split bush bearings

Answer: A

101. Which one of the following is considered a merit of the metal casting process?

- A. High-volume and low-skilled labor
- B. Internal porosity
- C. Dimensional variations
- D. High-tolerance, and smooth surfaces not possible

Answer: A

102. Which one of the following is **NOT** part of the gating systems in the sand casting process?

- A. Downsprue
- B. Riser
- C. Chaplet
- D. Runner

Answer: C

103. Which of the following material properties is **NOT** essential for metal forming?

- A. Proportional limit
- B. Young's Modulus
- C. Machinability
- D. Failure Strength

Answer: C

104. Which statement is **TRUE** about the chip formation process?

- A. The form of the chips is an important index of machining.
- B. Chip formation is quite simple process.
- C. Machining of brittle materials generally produces flat, curved or coiled continuous chips.
- D. Machining of ductile material produces discontinuous chips and mostly of irregular size and shape.

Answer: A

105. During the turning of a round metal workpiece on the CNC lathe machine, which force component is responsible for the estimation of power consumption?

- A. Axial, F_x
- B. Radial or transverse, F_y
- C. Tangential, F_z
- D. Resultant, R

Answer: C

106. Which of the following has a greater impact on the longitudinal strength of reinforced composites?

- A. Fiber orientation
- B. Fiber strength
- C. Fiber length
- D. Fiber diameter

Answer: B

107. The process carried out in powder metallurgy is as follows:

- i. Grading of powder
- ii. Preparation of powder
- iii. Compacting of powder
- iv. Sintering

The **CORRECT** sequence is

- A. i, ii, iii, iv
- B. i, iii, ii, iv
- C. ii, i, iii, iv
- D. iii, i, ii, iv

Answer: C

108. While drilling holes in steel plate by a 20 mm diameter HSS drill at a given feed, the tool life decreased from 40 min. to 24 min. when speed was raised from 250 rpm to 320 rpm. At what speed (rpm) the life of that drill under the same condition would be 30 min.?
- A. 267 rpm
B. 287 rpm
C. 277 rpm
D. 297 rpm

Answer: B

109. Choose the correct sequence to generate prototype
- A. 3D CAD data - CAD solid model - STL file - RP prototype
 - B. CAD solid model - 3D CAD data - RP prototype - STL file
 - C. STL file - 3D CAD data - CAD solid model - RP prototype
 - D. 3D CAD data - STL file - CAD solid model - RP prototype

Answer: A

110. Which one of the following is **WRONG** about non-traditional machining processes?
- A. Required when workpiece material is too hard, strong, or tough
 - B. Required when workpiece is too flexible to resist cutting forces or too difficult to clamp.
 - C. Required when part shape is very complex with internal or external profiles or small holes.
 - D. Primary method of material removal mechanisms is shearing.

Answer: D

111. In the case of gas welding, under which flame type is most welding performed?
- | | |
|--------------------|-------------------------|
| A. Oxidizing flame | C. Carburizing flame |
| B. Neutral flame | D. Oxy-carburized flame |

Answer: B

112. Which welding process allows a welding operation to be effectively completed without any physical contact being made between the electrode and the workpiece?
- | | |
|-----------------------|---------------------------|
| A. Gas Tungsten Arc | C. Flux-Cored Arc Welding |
| B. Plasma Arc Welding | D. Electro slag welding |

Answer: B

113. For successful spare parts management, it is critical to analyze the spare parts inventory based on various characteristics such as the frequency of issues, the annual consumption value, the criticality, the lead time, and the unit price in inventory analysis and selective control. One is wrong about these:
- A. FSN Analysis
B. ABC Analysis
C. RCF Analysis
D. VED Analysis

Answer: C

114.is concerned with proactively seeking the basic causes of facility/equipment failure.

- A. Maintenance planning
- B. Root cause failure analysis
- C. Fracture mechanics
- D. Excessive machine breakdown

Answer: B

115. One is **INCORRECT** about the failure mode of machinery's components.

- A. The specific characteristics of materials that result in the failure
- B. Evaluation of the materials' behavior using standard failure analysis practices typically identifies the failure mode.
- C. An elements of a component going through a separation due to imposed stresses.
- D. It may generally describe the way the failure occurs.

Answer: C

116. Maintenance planning is a comprehensive procedure that locates and addresses any potential problems ahead of time. It includes identifying and gathering the parts and/or tools required for jobs, making sure they are available and set out in the proper locations, having a planner write out instructions on how to complete a job, and even doing so before a job is assigned. As a result, the **CORRECT** major steps required for effective job of planning is.....

- A. Knowledge about equipment, job, available techniques, materials and facilities.
- B. Job investigation at site
- C. Preparation tools and facilities list indicating the needs of special tools, tackles and facilities needed.
- D. All

Answer: D

117. Planned maintenance is defined as "maintenance that is organized and carried out with forethought, control, and the use of records according to a predetermined plan." The task is completed with some thought, advance planning, record keeping, and control action when performing planned maintenance. One method differs from the following planned maintenance techniques.

- A. Preventive /Proactive maintenance
- B. Motivation and organization culture
- C. Predictive maintenance
- D. Corrective/Reactive maintenance.

Answer: B

118. It is known that after the spare parts have been coded and assigned stock location numbers, all users should be informed of and given the relevant codes and stock location numbers in the form of a spare parts catalogue. The information that should be included in the spare parts catalogue is.....

- A. Identification of spare parts and forecasting of spare parts requirement
- B. Spare parts codification plan and Spare part code
- C. Manufacturer's code & part number."
- D. B&C

Answer: D

119. A company is incurring 50,000 Birr of fixed cost to produce 5,000 units, and its variable cost per unit is 5 Birr. What is the average fixed cost?

- A. 5 Birr
- B. 10 Birr
- C. 15 Birr
- D. 8 Birr

Answer: B

120. Calculate compound interest for 10000 birr investment with 14% of interest rate for 6 years and interest is compounding half yearly.

- A. 22522 Birr
- B. 22222 Birr
- C. 12522 Birr
- D. 12222 Birr

Answer: A

121. Working capital cost is equal to

- A. Resource cost +Equipping cost
- B. Constructing cost +Equipping cost
- C. Resource cost +Operating cost
- D. Constructing cost +Operating cost

Answer: C

122. Time buffers can be add _____ bottlenecks in critical path.

- A. Between
- B. End of the
- C. At starting point
- D. Any place

Answer: A

123. Payment of Invoice can be prepared

- A. After Purchase order
- B. During follow up
- C. Before Purchase order
- D. After received materials

Answer: A

124. Autocorrelation factor is the relationship between

- A. Cyclical factors and trend line
- B. Previous three years data
- C. Past and future
- D. Average value

Answer: C